# Practice workbook · State the variables to control in an investigation

Bench investigations and investigations on people where the student lists factors to keep the same.

21 question parts: 1 worked-example retry and 20 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-NAME-CONTROL-VARIABLES

# 1. Question 2 · part a.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part a.ii only.

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**2** A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 $cm^3$ of each solution into a different test tube
- add 1.0 $cm^3$ of a competitive inhibitor and 1.0 $cm^3$ of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 $cm^3$ of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

2 .....

3 .....

# 1. Question 2 · part a.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part a.ii only.

(b) The table shows the student's results.

| Concentration of phenolphthalein phosphate solution in arbitrary units | Intensity of pink colour in arbitrary units |
|---|---|
| 0.05 | 0.20 |
| 0.10 | 0.30 |
| 0.20 | 0.47 |
| 0.40 | 0.65 |
| 0.60 | 0.79 |
| 0.80 | 0.91 |

(i) Plot the results of the investigation. (3)

(ii) Draw a curve of best fit. (1)

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## 1. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

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# 2. Question 2 · part c.ii · 2 marks

June 2019 Paper 1R

Further practice · Attempt part c.ii only.

**2** Humans need a balanced diet.

(a) Vitamins are needed in a balanced diet.

State four other components of a balanced diet.

**(2)**

1 .....

2 .....

3 .....

4 .....

(b) Explain why vitamin C is needed in a balanced diet.

**(2)**

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## 2. Question 2 · part c.ii · 2 marks

June 2019 Paper 1R

Further practice · Attempt part c.ii only.

(c) The vitamin C content of fruit juices varies.

(i) Design an investigation to compare the vitamin C content of fresh fruit juice and long-life fruit juice.

**(5)**

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(ii) State two variables that should be controlled in this investigation.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 11 marks)**

## 2. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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## 3. Question 9 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

**9** Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

**(3)**

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

**(3)**

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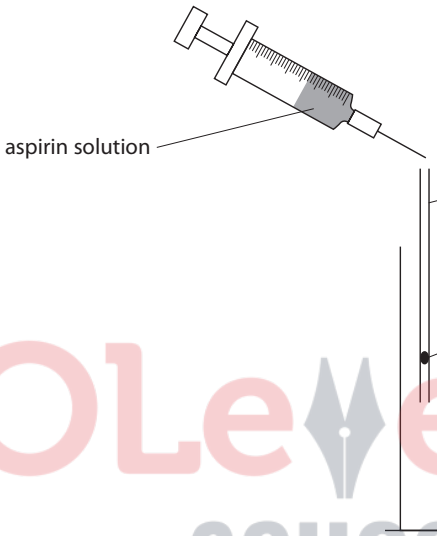
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## 3. Question 9 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.




The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

**(1)**

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(ii) Describe a suitable control test for this investigation.

**(1)**

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## 3. Question 9 · part b.i · 1 mark · Working and answer

Working and answer

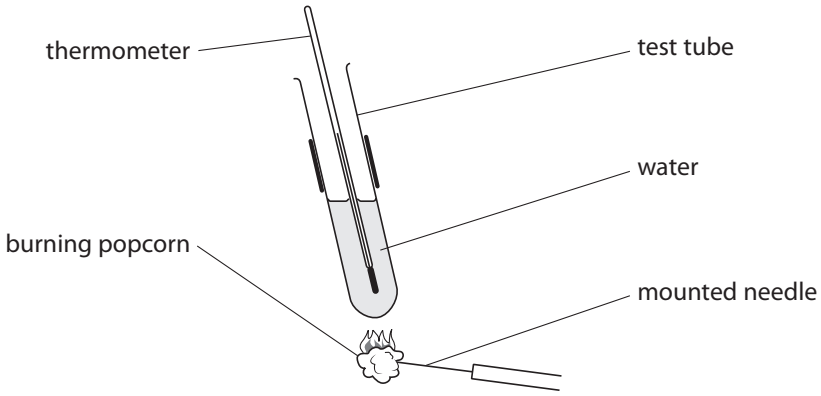
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# 4. Question 2 · part c · 4 marks

November 2020 Paper 1

Further practice · Attempt part c only.

**2** A student uses this apparatus to investigate the energy contained in popcorn.




This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

**(1)**

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(ii) Explain one safety precaution that the student should take during the investigation.

**(2)**

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## 4. Question 2 · part c · 4 marks

November 2020 Paper 1

Further practice · Attempt part c only.

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(b) The student uses this equation to calculate the energy released by the popcorn.

energy released = mass (g) × 4.2 × temperature rise (°C)

The initial temperature of the 12 g of water is 17 °C.

The maximum temperature of the water is 45 °C.

(i) Calculate the energy released when the student burns the popcorn.

Give your answer to 2 significant figures.

**(3)**

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energy released = ..... J

(ii) Explain why the calculated amount of energy released is not the same as the full amount of energy contained in the popcorn.

**(3)**

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## 4. Question 2 · part c · 4 marks

November 2020 Paper 1

Further practice · Attempt part c only.

(c) The student wants to compare the energy content of popcorn with the energy content of two other foods.

State four factors that the student should control to make sure the comparison is valid.

**(4)**

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3 .....

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4 .....

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**(Total for Question 2 = 13 marks)**

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## 4. Question 2 · part c · 4 marks · Working and answer

Working and answer

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## 5. Question 7 · part b.i · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

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**7** Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

**(2)**

(ii) Explain why chymosin is only needed in young humans.

**(2)**

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

**(3)**

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# 5. Question 7 · part b.i · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

| Number of bubbles of $CO_2$ | Time taken for chymosin to turn milk protein solid, in seconds | | | | |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 3 | Test 4 | Mean |
| 100 | 253 | 257 | 250 | 260 | 255 |
| 150 | 238 | 232 | 241 | 229 | |
| 200 | 216 | 214 | 219 | 211 | 215 |
| 250 | 208 | 202 | 212 | 198 | 205 |
| 300 | 210 | 200 | 199 | 311 | 203 |

(i) State three factors that the student should control.

**(3)**

1 .....

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2 .....

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3 .....

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(ii) Calculate the missing mean (average) time taken for 150 bubbles.

**(2)**

mean time taken = .....

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P 6 2 0 5 4 A 0 2 0 2 4

## 5. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

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# 6. Question 4 · part a.ii · 2 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

**4** Aspirin is the active ingredient used in some tablets to relieve pain.

Aspirin becomes acidic when it dissolves in the body. The faster aspirin dissolves, the faster it can relieve pain.

A student investigates if it is better to mix sucrose or starch with the active ingredient to make an aspirin tablet.

This is the student's method.

- add distilled water to three beakers, A, B and C
- add 1 aspirin tablet to each beaker
- add 5 g of sucrose to beaker B
- add 5 g of starch to beaker C
- measure the pH of the contents of each beaker at the start of the investigation
- measure the pH of the contents of each beaker every minute for 5 minutes

The table shows the student's results.

| Beaker | pH of contents of beaker | | | | | |
|---|---|---|---|---|---|---|
| | at start | 1 minute | 2 minutes | 3 minutes | 4 minutes | 5 minutes |
| A – aspirin | 7.6 | 7.1 | 6.6 | 6.2 | 5.8 | 5.4 |
| B – aspirin and sucrose | 7.2 | 6.7 | 6.3 | 5.7 | 5.2 | 4.7 |
| C – aspirin and starch | 7.1 | 7.2 | 7.3 | 7.3 | 7.4 | 7.4 |

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# 6. Question 4 · part a.ii · 2 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

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(a) (i) Plot a graph of the student's results.

Join the points with straight lines.

**(5)**



(ii) State two variables that should be controlled in this investigation.

**(2)**

1 .....

2 .....

(iii) Name the dependent variable in this investigation.

**(1)**

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## 6. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

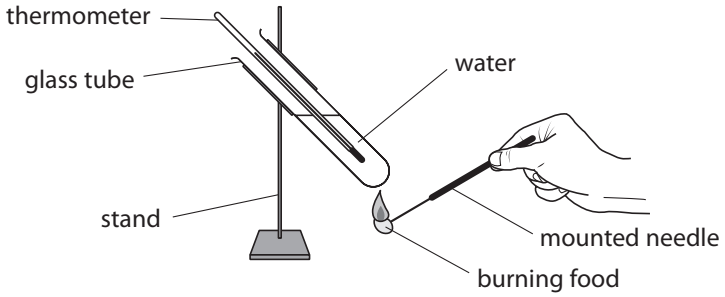
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# 7. Question 7 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

**7** A student uses this apparatus to investigate the energy content of different foods.




This is the student's method.

- add water to the glass tube
- measure the temperature of the water
- place a piece of food onto the mounted needle
- set the food alight by placing it into a Bunsen flame
- place the burning food under the glass tube
- measure the temperature of the water when the food stops burning

Repeat the method using different foods.

(a) (i) State three safety precautions that the student should take.

**(3)**

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2 .....

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3 .....

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# 7. Question 7 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

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(ii) State three factors that should be controlled during the investigation.

**(3)**

1 .....

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2 .....

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3 .....

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(iii) Explain why the energy content of a sample of food is likely to be higher than the energy of the same sample measured by this method.

**(5)**

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## 7. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

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# 8. Question 7 · part a.i · 3 marks

June 2021 Paper 2

Further practice · Attempt part a.i only.

**7** Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 $cm^3$ of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 $cm^3$ of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

| Petri dish | Diameter of bacterial growth in mm | | | | |
|---|---|---|---|---|---|
| | day 1 | day 2 | day 3 | day 4 | day 5 |
| A | 0 | 1.7 | 3.0 | 4.6 | 7.1 |
| B | 0 | 4.2 | 8.4 | 15.1 | 36.5 |
| C | 0 | 0 | 0 | 0 | 0 |

(a) (i) State three control variables in the student's investigation.

**(3)**

1 .....

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2 .....

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3 .....

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## 8. Question 7 · part a.i · 3 marks

June 2021 Paper 2

Further practice · Attempt part a.i only.

(ii) State the dependent variable in the student's investigation.

**(1)**

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(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

**(2)**

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

**(4)**

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

**(2)**

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## 8. Question 7 · part a.i · 3 marks · Working and answer

Working and answer

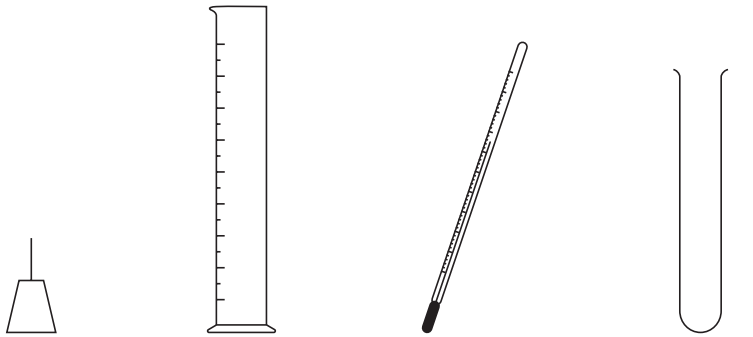
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# 9. Question 3 · part a.ii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.ii only.

**3** (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

**(5)**

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# 9. Question 3 · part a.ii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.ii only.

(ii) Name two variables that need to be controlled in the investigation.

**(2)**

1 .....

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2 .....

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(iii) Give two safety precautions needed for this investigation.

**(2)**

1 .....

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2 .....

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## 9. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

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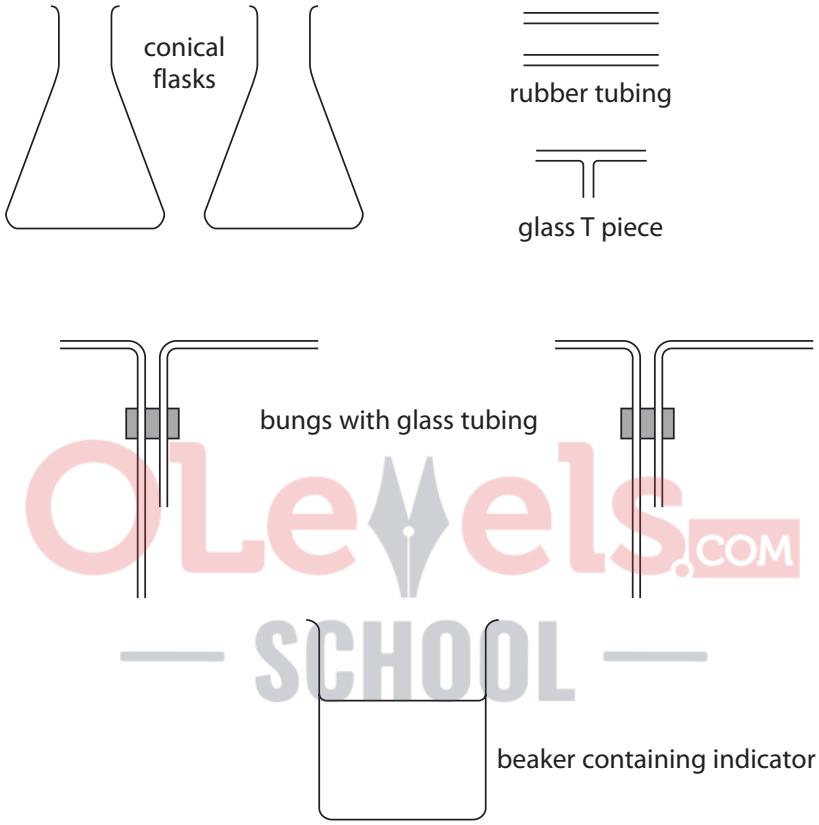
## 10. Question 1 · part e · 1 mark

June 2022 Paper 2R

Further practice · Attempt part e only.

**1** A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.




The apparatus is not drawn to scale.

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## 10. Question 1 · part e · 1 mark

June 2022 Paper 2R

Further practice · Attempt part e only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

**(3)**

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(b) Name an indicator that could be used in the investigation.

**(1)**

(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

**(4)**

## 10. Question 1 · part e · 1 mark

June 2022 Paper 2R

Further practice · Attempt part e only.

(d) State one safety precaution.

**(1)**

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(e) State one variable that should be controlled.

**(1)**

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**(Total for Question 1 = 10 marks)**

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Working and answer

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# 11. Question 6 · part a.ii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

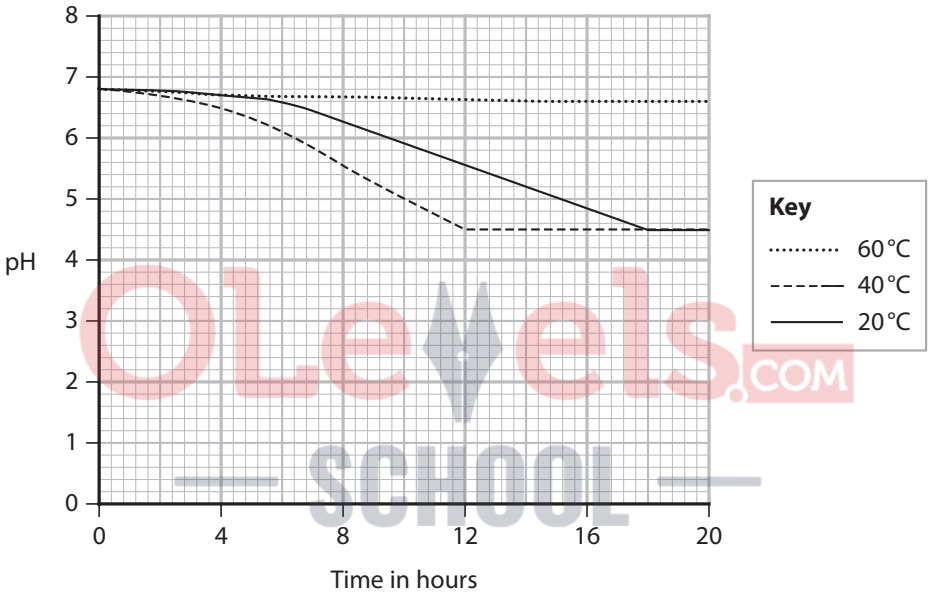
**6** Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.




(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

**(2)**

time = ..... hours

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## 11. Question 6 · part a.ii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

(ii) Give two variables that need to be controlled in this investigation.

**(2)**

1 .....

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2 .....

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(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

**(2)**

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(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

**(2)**

boiling

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cooling

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## 11. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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# 12. Question 3 · part b.ii · 2 marks

January 2023 Paper 2

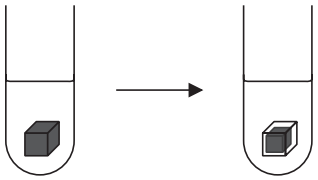
Further practice · Attempt part b.ii only.

**3** A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

**(3)**

surface area to volume ratio = .....

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

**(1)**

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

**(1)**

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## 12. Question 3 · part b.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(iv) Explain the effect of increasing the surface area to volume ratio on the time taken for the cube to become colourless.

**(2)**

(b) A student wants to investigate the effect of temperature on the rate of diffusion into the agar cube.

(i) Describe a method for this investigation using the apparatus shown in the diagram.

**(2)**

(ii) Give two variables the student would need to control in their investigation.

**(2)**

1 .....

2 .....

**(Total for Question 3 = 11 marks)**

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## 12. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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# 13. Question 2 · part b.iv · 1 mark

June 2023 Paper 2R

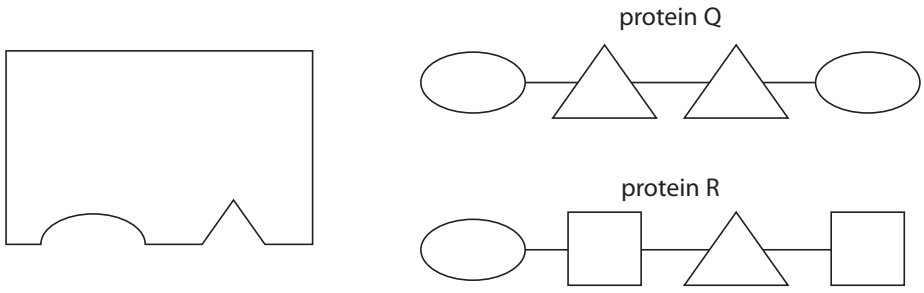
Further practice · Attempt part b.iv only.

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**2** (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.




**Enzyme P** **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

**(3)**

| amino acids | amylases | fatty acids | proteases |
|---|---|---|---|
| protein Q | protein R | starch | sugar |

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

**(1)**

.....

(iii) Name the molecule that codes for the production of enzyme P.

**(1)**

.....

# 13. Question 2 · part b.iv · 1 mark

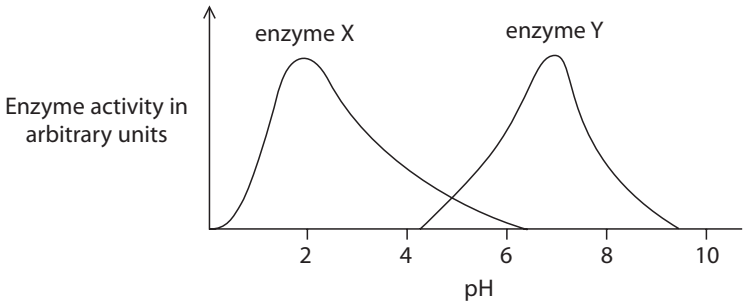
June 2023 Paper 2R

Further practice · Attempt part b.iv only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.




(i) Describe how the activity of enzyme X changes with pH.

**(3)**

(ii) Explain why no activity is shown for enzyme Y at pH2.

**(2)**

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## 13. Question 2 · part b.iv · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.iv only.

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(iii) Describe how the student could make the results of this investigation more reliable.

**(2)**

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.....

(iv) Name one control variable in the student's investigation.

**(1)**

.....

**(Total for Question 2 = 13 marks)**

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Working and answer

# 14. Question 7 · part a.i · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.i only.

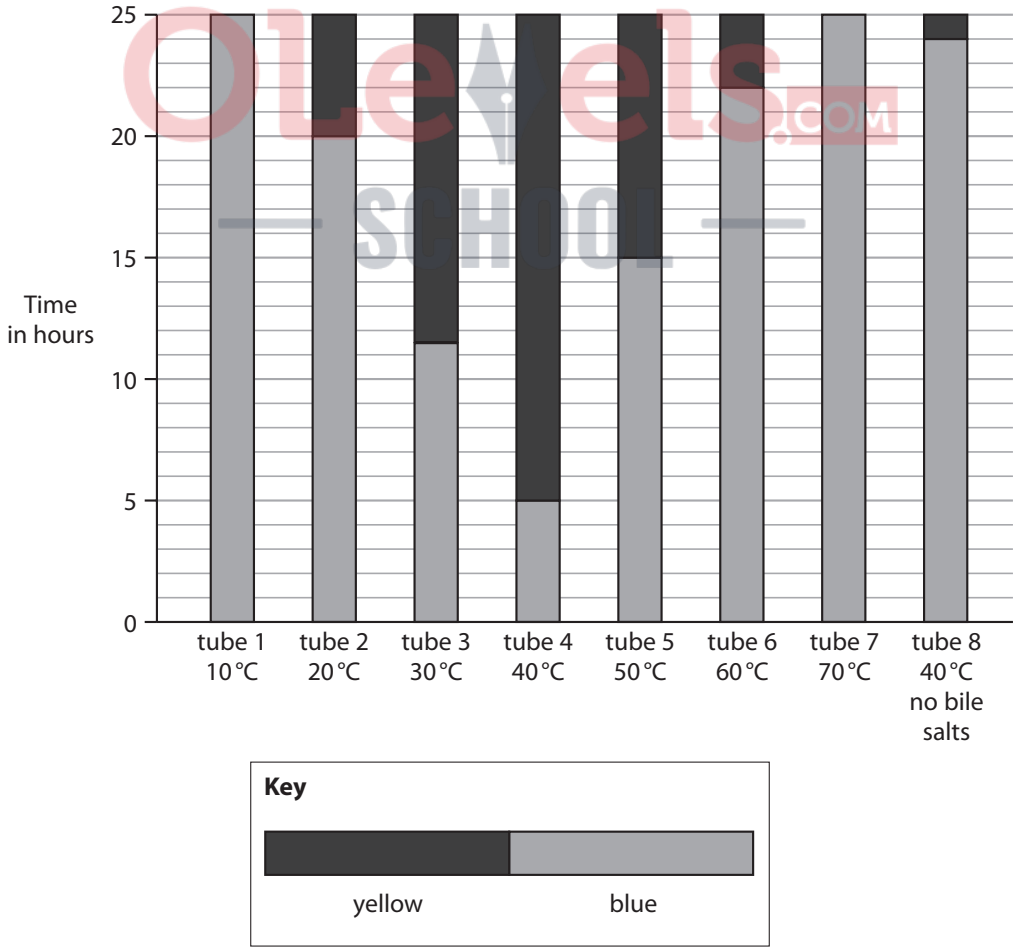
**7** A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.




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# 14. Question 7 · part a.i · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.i only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

**(3)**

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

**(1)**

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

**(3)**

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

**(2)**

time = ..... hours

(ii) Explain why the pH decreases during the 25 hours.

**(3)**

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## 14. Question 7 · part a.i · 3 marks · Working and answer

Working and answer

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# 15. Question 2 · part d · 2 marks

November 2024 Paper 1

Further practice · Attempt part d only.

**2** A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

| Time in minutes | pH of solution |
|---|---|
| 1 | 6.8 |
| 2 | 6.4 |
| 3 | 6.0 |
| 4 | 5.8 |
| 5 | 5.6 |
| 6 | 5.2 |
| 7 | 5.0 |
| 8 | 4.8 |
| 9 | 4.6 |
| 10 | 4.6 |

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## 15. Question 2 · part d · 2 marks

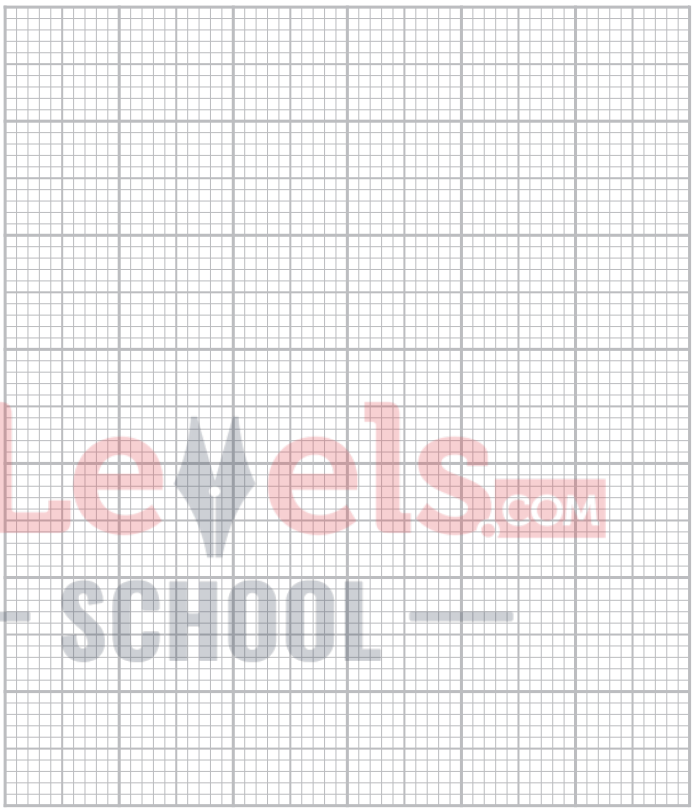
November 2024 Paper 1

Further practice · Attempt part d only.

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(a) (i) Plot a graph of the student's results.

**(3)**



(ii) Draw a line of best fit.

**(1)**

(iii) Give the trend shown by the graph.

**(1)**

.....

.....

## 15. Question 2 · part d · 2 marks

November 2024 Paper 1

Further practice · Attempt part d only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

**(3)**

(c) Name the independent variable in this investigation.

**(1)**

(d) Give two variables that the student should control in this investigation.

**(2)**

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

**(1)**

**(Total for Question 2 = 12 marks)**

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## 15. Question 2 · part d · 2 marks · Working and answer

Working and answer

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# 16. Question 4 · part b.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.iii only.

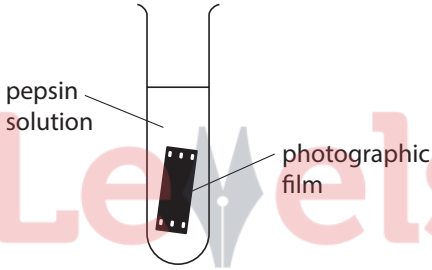
**4** Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin




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The table shows the student's results.

| Percentage pepsin concentration (%) | Time taken for film to become clear in seconds |
|---|---|
| 1 | 112 |
| 2 | 102 |
| 3 | 93 |
| 4 | 84 |
| 5 | 84 |
| 6 | 84 |

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# 16. Question 4 · part b.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.iii only.

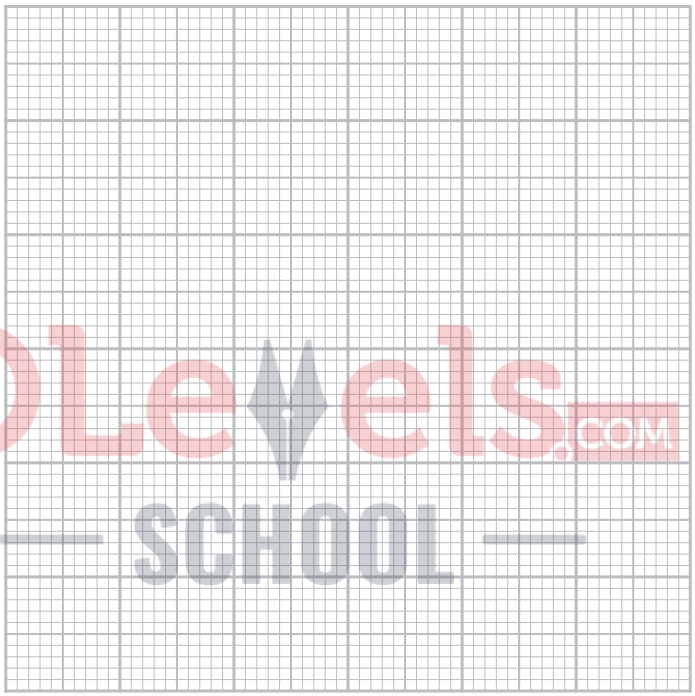
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(a) (i) Plot a graph of the results on the grid.

**(4)**

(ii) Draw a curve of best fit.

**(1)**



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(b) (i) Explain why the photographic film becomes clear.

**(3)**

## 16. Question 4 · part b.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.iii only.

(ii) Explain which would be the best pH to carry out this investigation.

**(2)**

(iii) State three variables that should be controlled during this investigation other than pH.

**(3)**

1 .....

2 .....

3 .....

(c) Suggest why the time taken for the film to become clear is the same for 4%, 5% and 6% concentrations of pepsin.

**(2)**

**(Total for Question 4 = 15 marks)**

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## 16. Question 4 · part b.iii · 3 marks · Working and answer

Working and answer

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# 17. Question 4 · part c.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.ii only.

**4** A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

| | | |
|---|---|---|
| 1.00 au = −7 % | 1.50 au = −10 % | 0.00 au (distilled water) = +8 % |
| | 0.25 au = +4 % | 0.50 au = −1 % |

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

**(4)**

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P 7 8 7 6 9 A 0 8 2 0

# 17. Question 4 · part c.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.ii only.

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(ii) Determine the median value for sugar concentration.

**(1)**

median value = .....

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

**(3)**

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mass of cylinder = ..... g

(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

**(4)**

concentration of sugar solution = ..... au

## 17. Question 4 · part c.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.ii only.

(c) (i) Explain why filter paper was wrapped around the potato cylinders at the beginning and end of the investigation.

**(2)**

(ii) The student kept the potato cylinders in the liquid for two hours.

State two other factors that should be kept constant during this investigation.

**(2)**

1 .....

2 .....

**(Total for Question 4 = 16 marks)**

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Working and answer

## 18. Question 2 · part c · 1 mark

January 2020 Paper 2

Further practice · Attempt part c only.

**2** A student records her pulse rate in beats per minute (bpm) after different activities.

These are the student's results.

- at rest 78 bpm
- walking 95 bpm
- jogging 138 bpm
- running 146 bpm

(a) Draw a table to show the student's results.

**(3)**

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## 18. Question 2 · part c · 1 mark

January 2020 Paper 2

Further practice · Attempt part c only.

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(b) Describe a method that the student could use to obtain these results.

**(4)**

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(c) State one factor that the student should control during the investigation.

**(1)**

(d) Name a body process, other than pulse rate, that increases during exercise.

**(1)**

**(Total for Question 2 = 9 marks)**

## 18. Question 2 · part c · 1 mark · Working and answer

Working and answer

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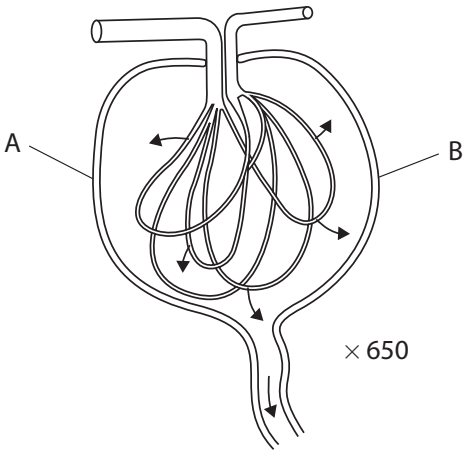
## 19. Question 1 · part b.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

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**1** (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).




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(i) Complete the diagram to show the missing parts of the kidney tubule.

**(3)**

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

**(3)**

width of Bowman's capsule = ..... mm



P 6 7 0 6 1 A 0 3 2 4

# 19. Question 1 · part b.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

**(2)**

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

**(1)**

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 $dm^3$ of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

| Time after drinking in hours | Volume of urine in $cm^3$ | |
|---|---|---|
| | day 1 (20 °C) | day 2 (30 °C) |
| 0.5 | 70 | 50 |
| 1.5 | 175 | 70 |
| 2.0 | 300 | 200 |
| 2.5 | 250 | 190 |
| 3.0 | 210 | 150 |
| 4.0 | 100 | 50 |

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## 19. Question 1 · part b.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

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(i) State two factors that should be controlled in the investigation.

**(2)**

1 .....

.....

2 .....

.....

(ii) The mean rate of urine production on day 1 is 276 $cm^3$ per hour.

Calculate the mean rate of urine production on day 2 during the 4 hour time period.

**(2)**

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mean rate of urine production on day 2 = ..... $cm^3$ per hour

(iii) Explain the difference between the mean rate on day 1 and the mean rate on day 2.

**(3)**

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**(Total for Question 1 = 16 marks)**

## 19. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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## 20. Question 5 · part a.ii · 3 marks

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

**5** A student uses this method to investigate the effect of exercise on the production of sweat.

- place a person in a room which has a temperature of 20 °C
- ask the person to remain inactive for 5 minutes
- measure the amount of sweat produced by a 25 $cm^2$ area of skin
- ask the person to exercise and measure the amount of sweat produced
- ask the person to rest for 10 minutes

Repeat the method at room temperatures of 22 °C, 24 °C, 26 °C, 28 °C and 30 °C

(a) (i) Explain why the person rests for 10 minutes before the method is repeated.

**(2)**

(ii) State three variables that should be controlled.

**(3)**

1 .....

2 .....

3 .....

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## 20. Question 5 · part a.ii · 3 marks

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

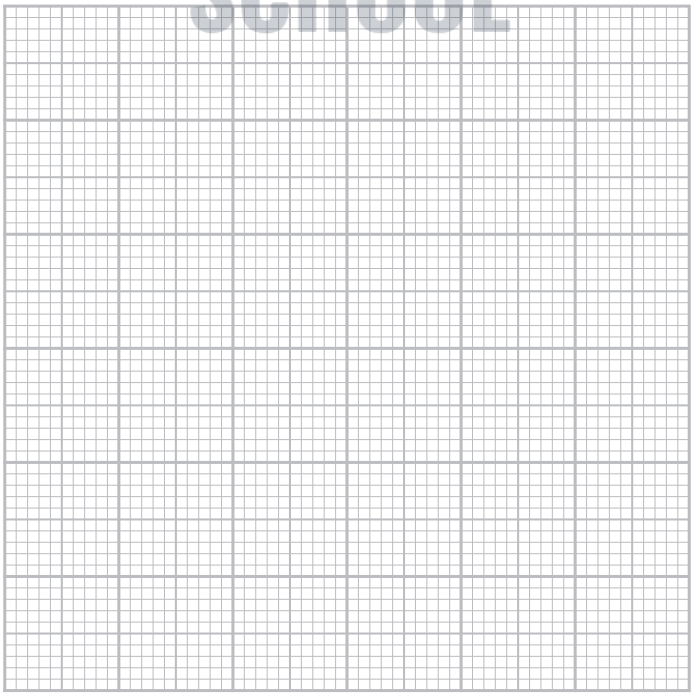
(b) The table shows the student's results.

| Room temperature in °C | Amount of sweat produced by 25 $cm^2$ area of skin in arbitrary units | |
|---|---|---|
| | While inactive | During exercise |
| 20 | 1 | 2 |
| 22 | 2 | 6 |
| 24 | 2 | 8 |
| 26 | 2 | 11 |
| 28 | 3 | 14 |
| 30 | 4 | 22 |

(i) Plot line graphs of the student's results.

Join the points with straight lines.

**(6)**



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P 7 2 5 8 6 A 0 1 2 2 0

## 20. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

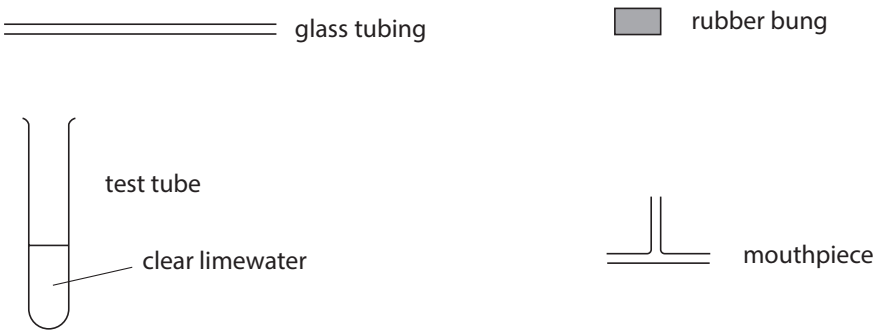
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# 21. Question 4 · part b.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.i only.

**4** (a) The diagram shows some scientific equipment.




(i) Draw a diagram to show how this equipment is set up to investigate the difference between the carbon dioxide concentration of inspired air and of expired air.

**(3)**

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# 21. Question 4 · part b.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.i only.

(ii) State one safety precaution in this investigation.

**(1)**

(b) A student investigates their breathing rate at rest, after gentle exercise and again after vigorous exercise.

The student repeats each exercise three times.

The student's mean results are shown in the table.

| Mean breathing rate in breaths per minute | | |
|---|---|---|
| **at rest** | **after gentle exercise** | **after vigorous exercise** |
| 14 | 19 | 50 |

(i) Give one control variable in this investigation.

**(1)**

(ii) Explain how the breathing rate after vigorous exercise benefits working muscles.

**(4)**

**(Total for Question 4 = 9 marks)**

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## 21. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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